Table 1. Grenada: External Debt Sustainability Framework, Baseline Scenario, 2015–2038 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 7/	Standard Deviation 7/	Projections									
	2015	2016	2017			2018	2019	2020	2021	2022	2023	2018-2023 Average	2028	2038	2024-2038 Average
External debt (nominal) 1/	**133.2**	**126.3**	**116.4**			**108.4**	**107.2**	**100.8**	**98.9**	**97.4**	**95.9**		**85.2**	**63.2**	
of which: public and publicly guaranteed (PPG)	61.4	56.9	48.0			47.1	43.8	39.2	38.9	39.0	39.2		35.8	33.1	
Change in external debt	-9.8	-6.9	-9.8			-8.1	-1.2	-6.4	-1.9	-1.5	-1.5		-1.9	-1.9	
Identified net debt-creating flows	-17.3	-12.8	-8.3			-5.1	-5.3	-5.5	-5.3	-5.0	-5.0		-2.5	2.0	
Non-interest current account deficit 2/	**1.4**	**0.9**	**4.5**	**11.8**	**10.3**	**5.8**	**6.1**	**5.4**	**5.3**	**5.6**	**5.6**		**8.0**	**11.1**	8.9
Deficit in balance of goods and services	-1.3	-3.7	-0.1			0.3	0.4	-0.4	-0.5	-0.3	-0.4		1.7	4.5	
Exports	58.0	56.1	56.3			56.3	56.2	56.0	55.9	55.9	55.8		54.4	52.3	
Imports	56.7	52.4	56.2			56.6	56.6	55.6	55.4	55.6	55.3		56.1	56.8	
Net current transfers (negative = inflow)	-0.7	1.2	1.0	-1.7	2.3	1.3	1.3	1.2	1.2	1.1	1.2		1.4	1.4	1.5
of which: official	0.8	1.4	1.2			1.5	1.5	1.5	1.5	1.5	1.5		1.5	1.5	
Other current account flows (negative = net inflow)	3.4	3.5	3.6			4.2	4.5	4.6	4.6	4.7	4.8		4.9	5.1	
Net FDI (negative = inflow)	**-8.9**	**-8.6**	**-8.5**	**-8.6**	**3.4**	**-8.6**	**-9.1**	**-9.1**	**-9.1**	**-9.1**	**-9.1**		**-9.1**	**-8.2**	-9.0
Endogenous debt dynamics 3/	**-9.8**	**-5.2**	**-4.3**			**-2.3**	**-2.3**	**-1.8**	**-1.5**	**-1.5**	**-1.5**		**-1.4**	**-0.9**	
Contribution from nominal interest rate	2.5	2.3	2.2			1.6	1.3	1.2	1.2	1.0	1.0		0.9	0.8	
Contribution from real GDP growth	-8.4	-4.6	-5.4			-3.9	-3.7	-3.0	-2.6	-2.6	-2.5		-2.3	-1.7	
Contribution from price and exchange rate changes	-3.8	-2.8	-1.1			...	...	...	...	...	...		...	...	
Residual (3-4) 4/	**7.5**	**5.9**	**-1.5**			**-3.0**	**4.1**	**-0.9**	**3.4**	**3.5**	**3.5**		**0.6**	**-3.9**	
of which: exceptional financing	-3.1	-1.5	-1.1			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 5/	...	...	108.6			99.1	98.0	92.8	90.7	88.9	87.2		77.0	55.3	
In percent of exports	...	...	192.7			175.9	174.4	165.6	162.2	159.1	156.4		141.5	105.8	
PV of PPG external debt	**...**	**...**	**40.1**			**37.8**	**34.7**	**31.1**	**30.6**	**30.5**	**30.4**		**27.5**	**25.3**	
In percent of exports	**...**	**...**	**71.3**			**67.0**	**61.6**	**55.6**	**54.8**	**54.6**	**54.6**		**50.6**	**48.3**	
In percent of government revenues	**...**	**...**	**172.3**			**165.0**	**154.9**	**139.2**	**137.4**	**137.0**	**137.2**		**123.9**	**113.4**	
Debt service-to-exports ratio (in percent)	**7.8**	**9.7**	**10.6**			**8.1**	**7.1**	**6.5**	**6.3**	**6.2**	**6.4**		**5.4**	**5.0**	
PPG debt service-to-exports ratio (in percent)	**7.4**	**9.4**	**10.3**			**7.8**	**6.8**	**6.2**	**6.0**	**5.9**	**6.1**		**5.2**	**4.9**	
PPG debt service-to-revenue ratio (in percent)	**20.3**	**23.0**	**24.8**			**19.1**	**17.0**	**15.5**	**15.0**	**14.9**	**15.3**		**12.8**	**11.4**	
Total gross financing need (Millions of U.S. dollars)	-30.0	-23.3	21.9			21.3	13.1	-0.5	-4.3	0.1	0.5		35.6	175.7	
Non-interest current account deficit that stabilizes debt ratio	11.2	7.9	14.4			13.9	7.3	11.8	7.2	7.1	7.1		9.8	13.0	
Key macroeconomic assumptions															
Real GDP growth (in percent)	6.4	3.7	4.5	2.9	3.1	3.6	3.6	2.9	2.7	2.7	2.7	3.0	2.7	2.7	2.7
GDP deflator in US dollar terms (change in percent)	2.8	2.2	0.9	1.8	1.4	2.7	2.2	2.2	2.2	2.2	2.2	2.3	2.2	2.2	2.2
Effective interest rate (percent) 6/	1.9	1.8	1.8	1.6	0.2	1.5	1.3	1.2	1.2	1.1	1.1	1.2	1.1	1.3	1.1
Growth of exports of G&S (US dollar terms, in percent)	5.5	2.5	5.8	23.2	46.5	6.4	5.7	4.8	4.8	5.0	4.7	5.2	4.5	4.7	4.6
Growth of imports of G&S (US dollar terms, in percent)	5.3	-2.2	13.2	8.5	9.5	7.2	5.9	3.2	4.6	5.5	4.5	5.1	5.1	5.1	5.2
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	34.0	49.8	22.8	28.2	27.4	27.0	31.5	27.3	25.8	27.0
Government revenues (excluding grants, in percent of GDP)	21.2	22.8	23.3			22.9	22.4	22.4	22.3	22.3	22.2		22.2	22.3	22.2
Aid flows (in Millions of US dollars) 8/	32.3	37.0	28.9			64.4	54.1	35.7	43.8	44.3	44.8		43.6	48.7	
of which: Grants	32.3	37.0	28.9			34.4	35.1	35.7	36.2	36.7	37.2		36.0	41.0	
of which: Concessional loans	0.0	0.0	0.0			30.0	19.0	0.0	7.6	7.6	7.6		7.6	7.6	
Grant-equivalent financing (in percent of GDP) 9/	...	...	...			5.7	3.8	2.7	3.7	3.7	3.7		2.8	2.1	2.5
Grant-equivalent financing (in percent of external financing) 9/	...	...	...			51.3	78.9	100.0	56.8	53.7	52.5		53.2	47.9	52.1
Memorandum items:															
Nominal GDP (Millions of US dollars)	997.0	1056.2	1113.3			1185.0	1254.3	1319.1	1385.6	1454.8	1527.3		1948.7	3173.0	
Nominal dollar GDP growth	9.4	5.9	5.4			6.4	5.8	5.2	5.0	5.0	5.0	5.4	5.0	5.0	5.0
PV of PPG external debt (in Millions of US dollars)			446.9			447.3	434.7	410.6	424.2	444.0	464.9		536.7	802.0	
(PVt-PVt-1)/GDPt-1 (in percent)						0.0	-1.1	-1.9	1.0	1.4	1.4	0.2	1.0	1.0	1.0
Gross workers' remittances (Millions of US dollars)	28.8	29.0	30.3			31.8	36.1	40.2	44.5	48.9	50.8		61.2	88.8	
PV of PPG external debt (in percent of GDP + remittances)	...	...	39.1			36.8	33.7	30.2	29.7	29.5	29.5		26.7	24.6	
PV of PPG external debt (in percent of exports + remittances)	...	...	68.0			64.0	58.6	52.7	51.8	51.5	51.5		47.9	45.9	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	9.8			7.4	6.4	5.9	5.6	5.6	5.7		4.9	4.6	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ The significant decline in projections for the non-interest CA balance relative to the previous DSA (Annex II, ECF Sixth Review, May 2017), is a result of the transition from BPM5 to BPM6 methodology.

3/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

4/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

5/ Assumes that PV of private sector debt is equivalent to its face value.

6/ Current-year interest payments divided by previous period debt stock.

7/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

8/ Defined as grants, concessional loans, and debt relief.

9/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

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Table 2. Grenada: Public Sector Debt Sustainability Framework, Baseline Scenario, 2015–2038

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate							Projections										
	2015	2016	2017	Average 5/	Standard Deviation 5/	2018	2019	2020	2021	2022	2023	2018-23 Average	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2024-38 Average
Public sector debt 1/	90.1	82.0	70.8			65.0	59.6	52.9	51.3	50.7	50.3		50.2	50.5	50.7	50.9	51.1	51.3	51.1	52.5	52.5	52.4	52.4	
of which: foreign-currency denominated	61.4	56.9	48.0			47.1	43.8	39.2	38.9	39.0	39.2		35.8	35.4	34.9	34.5	34.1	33.7	33.2	34.2	33.8	33.5	33.1	
Change in public sector debt	-11.7	-8.1	-11.2			-5.8	-5.4	-6.7	-1.6	-0.6	-0.4		0.3	0.2	0.2	0.2	0.2	0.2	-0.1	1.4	0.0	0.0	0.0	
Identified debt-creating flows	-13.2	-8.9	-12.4			-8.3	-7.2	-7.0	-4.7	-3.3	-2.8		-0.1	0.0	0.0	-0.1	-0.1	-0.1	-0.4	-0.3	-0.4	-0.4	-0.4	
Primary deficit	-2.1	-5.2	-5.7	0.2	3.6	-5.6	-5.3	-5.5	-3.6	-2.1	-1.6	-4.0	1.1	1.1	1.1	1.1	1.1	1.1	0.8	0.8	0.8	0.8	0.8	0.9
Revenue and grants	24.5	26.4	25.9			25.8	25.2	25.1	24.9	24.8	24.6		24.1	24.0	24.0	23.9	23.8	23.8	23.8	23.7	23.7	23.6	23.6	
of which: grants	3.2	3.5	2.6			2.9	2.8	2.7	2.6	2.5	2.4		1.8	1.8	1.7	1.7	1.6	1.5	1.5	1.4	1.4	1.3	1.3	
Primary (noninterest) expenditure	22.3	21.2	20.1			20.2	19.9	19.6	21.3	22.7	23.0		25.2	25.1	25.1	25.0	24.9	24.9	24.5	24.5	24.4	24.4	24.3	
Automatic debt dynamics	-5.4	-2.2	-1.5			-2.2	-1.9	-1.5	-1.1	-1.2	-1.2		-1.2	-1.2	-1.1	-1.2	-1.2	-1.2	-1.1	-1.1	-1.2	-1.2	-1.2	
Contribution from interest rate/growth differential	-4.3	-1.6	-2.0			-2.0	-2.2	-1.4	-1.1	-1.2	-1.2		-0.4	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4	-0.5	-0.5	
of which: contribution from average real interest rate	1.8	1.6	1.5			0.4	0.1	0.3	0.3	0.2	0.2		0.9	0.9	1.0	0.9	0.9	0.9	1.0	1.0	1.0	0.9	0.9	
of which: contribution from real GDP growth	-6.2	-3.2	-3.5			-2.5	-2.2	-1.7	-1.4	-1.4	-1.3		-1.3	-1.3	-1.3	-1.3	-1.4	-1.4	-1.4	-1.4	-1.4	-1.4	-1.4	
Contribution from real exchange rate depreciation	-1.1	-0.5	0.5			-0.1	0.3	-0.1	-0.1	0.0	0.0		...	...	...	...	...	...	...	...	...	...	...	
Other identified debt-creating flows	-5.7	-1.6	-5.2			-0.5	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Debt relief (HIPC and other)	-5.7	-1.0	-5.2			-0.5	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	-0.6	0.1			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Residual, including asset changes	1.6	0.8	1.2			2.5	1.8	0.3	3.1	2.7	2.4		0.3	0.3	0.3	0.3	0.3	0.3	0.3	1.7	0.3	0.4	0.4	
Other Sustainability Indicators																								
PV of public sector debt	...	...	62.9			55.7	50.5	44.9	43.0	42.2	41.6		41.9	42.2	42.5	42.8	43.1	43.4	43.3	44.4	44.5	44.5	44.6	
of which: foreign-currency denominated	...	...	40.1			37.8	34.7	31.1	30.6	30.5	30.4		27.5	27.1	26.7	26.4	26.1	25.9	25.4	26.2	25.8	25.6	25.3	
of which: external	...	...	40.1			37.8	34.7	31.1	30.6	30.5	30.4		27.5	27.1	26.7	26.4	26.1	25.9	25.4	26.2	25.8	25.6	25.3	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	...	...	...	...	...	...	...	...	...	
Gross financing need 2/	17.0	15.5	7.8			6.0	6.5	5.5	7.0	8.1	8.3		9.3	9.1	8.6	8.1	8.0	7.8	7.4	7.3	7.2	7.0	6.9	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	242.9			216.0	200.4	179.0	172.9	170.2	168.7		174.2	175.7	177.3	179.1	180.7	182.2	182.5	187.4	188.0	188.5	189.0	
PV of public sector debt-to-revenue ratio (in percent)	...	...	269.9			243.4	225.5	200.6	193.2	189.4	187.3		188.7	189.8	191.0	192.4	193.7	194.9	194.7	199.5	199.7	199.9	200.0	
of which: external 3/	...	...	172.3			165.0	154.9	139.2	137.4	137.0	137.2		123.9	121.9	120.2	118.8	117.5	116.2	114.3	117.5	116.0	114.7	113.4	
Debt service-to-revenue and grants ratio (in percent) 4/	78.5	78.6	52.1			45.1	46.8	43.8	42.6	41.1	40.4		33.7	33.2	31.4	29.4	28.9	28.4	28.0	27.5	27.0	26.5	26.0	
Debt service-to-revenue ratio (in percent) 4/	90.4	90.7	58.0			50.8	52.7	49.1	47.5	45.8	44.8		36.5	35.9	33.8	31.6	30.9	30.4	29.9	29.3	28.6	28.1	27.5	
Primary deficit that stabilizes the debt-to-GDP ratio	9.5	2.9	5.5			0.1	0.1	1.2	-2.0	-1.5	-1.2		0.9	0.9	0.9	0.9	0.9	0.9	0.9	-0.6	0.8	0.8	0.8	
Key macroeconomic and fiscal assumptions																								
Real GDP growth (in percent)	6.4	3.7	4.5	2.9	3.1	3.6	3.6	2.9	2.7	2.7	2.7	3.0	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7
Average nominal interest rate on forex debt (in percent)	3.7	3.6	3.7	2.9	0.8	3.1	2.5	2.4	2.6	2.4	2.3	2.6	2.2	2.3	2.3	2.3	2.3	2.3	2.4	2.5	2.3	2.3	2.3	2.3
Average real interest rate on domestic debt (in percent)	0.6	0.8	2.1	2.2	1.4	0.5	1.0	1.0	0.9	0.9	0.9	0.8	1.0	1.0	1.0	1.0	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.0
Real exchange rate depreciation (in percent, + indicates depreciation)	-1.6	-0.9	0.9	-0.2	1.4	-0.3	..	..	..	..	..	...	...	...	...	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	2.8	2.2	0.9	1.8	1.4	2.7	2.2	2.2	2.2	2.2	2.2	2.3	2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2
Growth of real primary spending (deflated by GDP deflator, in percent)	-7.3	-1.7	-0.5	-1.2	2.5	3.6	2.1	1.3	12.0	9.4	4.1	5.4	4.0	2.4	2.4	2.4	2.5	2.5	1.3	2.5	2.6	2.6	2.5	3.1
Grant element of new external borrowing (in percent)	..	..	..	...	...	34.0	49.8	22.8	28.2	27.4	27.0	31.5	27.3	27.1	27.2	27.4	27.1	26.8	26.8	25.3	26.2	26.0	25.8	27.0

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

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Table 3. Grenada: Sensitivity Analysis for Key Indicators of Publicly Guaranteed External Debt, 2018–2038

(In percent)

	Projections							
	2018	2019	2020	2021	2022	2023	**2028**	2038
PV of debt-to GDP ratio								
Baseline	38	35	31	31	31	30	**28**	25
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2018-2038 1/	38	40	41	46	51	56	**67**	58
A2. New public sector loans on less favorable terms in 2018-2038 2	38	35	32	32	33	34	**34**	38
A3. Alternative Scenario: Natural Disaster	38	41	35	35	35	35	**31**	29
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2019-2020	38	36	33	33	33	33	**29**	27
B2. Export value growth at historical average minus one standard deviation in 2019-2020 3/	38	46	62	61	61	61	**49**	32
B3. US dollar GDP deflator at historical average minus one standard deviation in 2019-2020	38	35	32	32	32	32	**29**	26
B4. Net non-debt creating flows at historical average minus one standard deviation in 2019-2020 4/	38	37	36	35	35	35	**31**	26
B5. Combination of B1-B4 using one-half standard deviation shocks	38	39	41	40	40	40	**35**	28
B6. One-time 30 percent nominal depreciation relative to the baseline in 2019 5/	38	49	44	43	43	43	**39**	36
PV of debt-to-exports ratio								
Baseline	67	62	56	55	55	55	**51**	48
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2018-2038 1/	67	70	74	82	91	100	**124**	111
A2. New public sector loans on less favorable terms in 2018-2038 2	67	63	57	57	59	61	**63**	73
A3. Alternative Scenario: Natural Disaster	67	73	62	62	62	63	**57**	55
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2019-2020	67	62	56	55	55	55	**51**	48
B2. Export value growth at historical average minus one standard deviation in 2019-2020 3/	67	113	208	206	205	204	**170**	115
B3. US dollar GDP deflator at historical average minus one standard deviation in 2019-2020	67	62	56	55	55	55	**51**	48
B4. Net non-debt creating flows at historical average minus one standard deviation in 2019-2020 4/	67	66	64	64	63	63	**57**	50
B5. Combination of B1-B4 using one-half standard deviation shocks	67	70	76	75	75	75	**67**	57
B6. One-time 30 percent nominal depreciation relative to the baseline in 2019 5/	67	62	56	55	55	55	**51**	48
PV of debt-to-revenue ratio								
Baseline	165	155	139	137	137	137	**124**	113
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2018-2038 1/	165	177	185	207	228	251	**303**	261
A2. New public sector loans on less favorable terms in 2018-2038 2	165	158	142	144	148	152	**155**	172
A3. Alternative Scenario: Natural Disaster	165	184	155	155	157	159	**140**	129
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2019-2020	165	161	149	147	147	147	**133**	121
B2. Export value growth at historical average minus one standard deviation in 2019-2020 3/	165	206	277	275	273	273	**222**	143
B3. US dollar GDP deflator at historical average minus one standard deviation in 2019-2020	165	158	144	142	142	142	**128**	117
B4. Net non-debt creating flows at historical average minus one standard deviation in 2019-2020 4/	165	166	161	159	159	159	**139**	118
B5. Combination of B1-B4 using one-half standard deviation shocks	165	173	183	181	180	180	**156**	128
B6. One-time 30 percent nominal depreciation relative to the baseline in 2019 5/	165	219	197	195	194	194	**175**	161

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